

Fructose intake at current levels in the United States may cause gastrointestinal distress in normal adults.

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OBJECTIVE: Fructose intake has increased considerably in the United States, primarily as a result of increased consumption of high-fructose corn syrup, fruits and juices, and crystalline fructose. The purpose was to determine how often fructose, in amounts commonly consumed, would result in malabsorption and/or symptoms in healthy persons. DESIGN: Fructose absorption was measured using 3-hour breath hydrogen tests and symptom scores were used to rate subjective responses for gas, borborygmus, abdominal pain, and loose stools. SUBJECTS/SETTING: The study included 15 normal, free-living volunteers from a medical center community and was performed in a gastrointestinal specialty clinic. INTERVENTION: Subjects consumed 25- and 50-g doses of crystalline fructose with water after an overnight fast on separate test days. MAIN OUTCOME MEASURES: Mean peak breath hydrogen, time of peak, area under the curve (AUC) for breath hydrogen and gastrointestinal symptoms were measured during a 3-hour period after subjects consumed both 25- and 50-g doses of fructose. STATISTICAL ANALYSES: Differences in mean breath hydrogen, AUC, and symptom scores between doses were analyzed using paired t tests. Correlations among peak breath hydrogen, AUC, and symptoms were also evaluated. RESULTS: More than half of the 15 adults tested showed evidence of fructose malabsorption after 25 g fructose and greater than two thirds showed malabsorption after 50 g fructose. AUC, representing overall breath hydrogen response, was significantly greater after the 50-g dose. Overall symptom scores were significantly greater than baseline after each dose, but scores were only marginally greater after 50 g than 25 g. Peak hydrogen levels and AUC were highly correlated, but neither was significantly related to symptoms. CONCLUSIONS: Fructose, in amounts commonly consumed, may result in mild gastrointestinal distress in normal people. Additional study is warranted to evaluate the response to fructose-glucose mixtures (as in high-fructose corn syrup) and fructose taken with food in both normal people and those with gastrointestinal dysfunction. Because breath hydrogen peaks occurred at 90 to 114 minutes and were highly correlated with 180-minute breath hydrogen AUC, the use of peak hydrogen measures may be considered to shorten the duration of the exam.